

I am an applied macroeconomist studying how individual-level behavior shapes the transmission of macroeconomic shocks and policy interventions. My research focuses on two related frictions that are central to macroeconomic policy: labor-market frictions and informational frictions. The first strand studies labor-market frictions, especially downward nominal wage rigidity (DNWR), and their implications for unemployment, fiscal policy, monetary policy, and welfare. The second strand studies informational frictions, focusing on how households form expectations, how they interpret economic information, and how communication shapes policy transmission. Together, these frictions influence how macroeconomic shocks propagate through the economy and why the effects of policy interventions differ across economic environments. Methodologically, my work combines microdata, survey experiments, laboratory experiments, and structural macroeconomic models. Four of my papers have been published in leading economics journals, and I have three additional working papers, including one under revision and two under review. My work has received 218 Google Scholar citations, including 124 since 2024, indicating a growing scholarly trajectory. In what follows, I summarize the main contributions of my research.

1. Downward Nominal Wage Rigidity and Macroeconomic Stabilization

Wage adjustment is an important channel through which macroeconomic shocks are transmitted to employment, inflation, and welfare. When labor demand falls, the extent to which wages can adjust downward affects whether firms respond through lower wages, lower employment, or both. For this reason, nominal wage rigidity has long been an important ingredient in macroeconomic models of business cycles and stabilization policy. Yet the policy implications of wage rigidity depend critically on how it is modeled. A model in which wages are symmetrically sticky can generate very different conclusions from a model in which wages are especially difficult to cut.

My work on downward nominal wage rigidity (DNWR) starts with a measurement and model-selection question: which form of wage rigidity is supported by individual-level wage data, and which wage-setting models best match those patterns? Although DNWR is widely used as a theoretical mechanism, measuring it in the data is challenging. Much of the empirical literature has focused on the spike at zero (Daly and Hobijn 2014) - the share of workers with no nominal wage change from one year to the next - but the spike at zero alone does not reveal how the rest of the wage-change distribution responds to labor-market conditions. In my sole-authored paper, “[1] Spike at Zero,” I use individual-level wage data from two nationally representative U.S. surveys (the CPS and the SIPP) to study the full distribution of wage changes over the business cycle. I show that downturns are associated with a substantially larger increase in the spike at zero than in wage cuts, both in aggregate U.S. data and across U.S. states. This excess cyclicalness of the spike at zero relative to wage cuts provides a sharper test of DNWR than the spike at zero alone.

I then use these empirical moments to evaluate heterogeneous-agent models with alternative wage-setting mechanisms, including flexible wages, Calvo rigidity, long-term contracts, menu costs, and DNWR. The results show that models incorporating DNWR are most consistent with the observed shape and cyclicity of the wage-change distribution. This comparison clarifies why the asymmetry of wage adjustment matters for understanding labor-market responses to economic shocks.

The paper makes two broader contributions. First, it provides new micro-level evidence that DNWR is the wage-setting mechanism most consistent with individual-level wage changes over the business cycle. Second, it develops a state-level measure of the extent to which DNWR binds. This measure makes it possible to study how variation in wage rigidity affects shock propagation and the effectiveness of stabilization policy, and it has subsequently been used in research on trade shocks (Rodríguez-Clare et al. (2026)), state-level recoveries from recessions (Ahn and Eo (2025)), and government spending shocks (Jo and Zubairy (2025)). By making downward wage rigidity measurable at the regional level, the paper contributes not only to the study of wage rigidity itself, but also to a broader research agenda on how shocks spread through the economy and how policy can stabilize employment. This paper was presented at Federal Reserve Bank of Chicago and Federal Reserve Bank of St. Louis and published in *Journal of International Economics*.

Building on this empirical foundation, my joint work with Sarah Zubairy (Texas A&M), “[2] Multiplier,” studies how DNWR affects the state dependence of government spending multipliers. The literature has long debated whether fiscal policy is more effective in recessions than in normal times (see, for example, Auerbach and Gorodnichenko (2012), Nakamura and Steinsson (2014), and Ramey and Zubairy (2018)). Our paper helps explain why prior studies have found mixed evidence: recessions differ in their underlying sources, and this distinction determines whether DNWR has real effects on labor-market adjustment. Once we distinguish demand-driven recessions from supply-driven recessions, the state dependence of government spending multipliers becomes clearer.

The key mechanism is that DNWR constrains nominal wage adjustment, while labor-market outcomes depend on real wages. In a demand-driven recession, inflation falls, preventing real wages from adjusting downward and generating inefficiently high unemployment. Under these conditions, government spending is especially effective because it raises labor demand while limiting crowding out through higher real wages and interest rates. In contrast, in a supply-driven recession, inflation rises, helping real wages adjust downward even when nominal wages are constrained. As a result, DNWR has weaker employment effects, and government spending multipliers are smaller.

The paper contributes to both the theoretical and empirical literature on state-dependent fiscal policy. Theoretically, it shows that the interaction between unemployment, inflation, and DNWR is central for understanding when government spending is most effective. Empirically, using both U.S. historical time-series data and state-level variation in government spending and DNWR, we show that spending multipliers are larger in low-inflation recessions, especially when DNWR binds most. These findings suggest that policymakers should consider not only the amount of slack in the economy, but also the source of the downturn and the inflation response when using government

spending as a stabilization tool. This project was supported by a Texas A&M T3 grant, presented at NBER, and published in *American Economic Journal: Macroeconomics*. The paper has attracted attention because it helps reconcile mixed evidence on the effectiveness of government spending and provides policy-relevant guidance on when fiscal stimulus is most effective.

My more recent work extends this agenda to currency unions. In “[3] **Currency Union**,” joint with Joonseok Oh (University of Southampton), I study the welfare implications of wage rigidity in countries that share a common currency. Currency unions are a natural setting for studying wage rigidity because member countries cannot use independent monetary policy to respond to local economic shocks. Existing research (Galí and Monacelli (2016)) has suggested that greater wage rigidity can improve welfare in such a monetary regime by stabilizing wage and price inflation. In contrast, our paper shows that this conclusion changes once DNWR is incorporated.

Using evidence from European firms, we document that wage freezes are widespread and more countercyclical than wage cuts, consistent with the evidence in “[1] Spike at Zero.” We then develop a currency-union model with DNWR and show that when downward wage adjustment is constrained, employment fluctuations become the main source of welfare loss. As a result, greater wage rigidity can worsen welfare, even in a currency union. More broadly, this project shows that the welfare consequences of wage rigidity are state-dependent and that incorporating DNWR can fundamentally change conclusions about the welfare effects of wage rigidity when monetary policy is constrained, such as in currency unions or at the zero lower bound.

A complementary strand of my research examines how technological change affects prices and consumer welfare through the lens of search and geographic frictions—a theme that connects to my broader interest in how micro-level frictions shape aggregate outcomes. In joint work with Misaki Matsumura (Deloitte Tax LLP) and David Weinstein (Columbia University), “[7] **E-Commerce**,” I study how the rise of retail e-commerce affected prices and welfare in Japan. Using detailed Japanese price and expenditure data, we show that e-commerce lowered relative inflation rates for goods sold intensively online and accelerated price convergence across cities. These findings suggest that e-commerce reduces search and geographic frictions, limiting the ability of local retailers to charge different prices across markets. We also estimate that the expansion of e-commerce lowered variety-adjusted prices by approximately 0.9 percent on average, generating measurable consumer welfare gains. A key contribution of the paper is to overcome important measurement and identification challenges by using long time-series data on prices and retail-channel expenditure shares, together with historical catalog sales as an instrument for e-commerce intensity. The paper was presented at the NBER Japan Project Meeting, published in the *Review of Economics and Statistics*, and is my most cited work to date. Its findings are increasingly relevant as e-commerce has become a larger part of shopping and retail activity.

2. Expectations, Belief Formation, and Policy Transmission

The second branch of my research studies how households form expectations and how those expectations shape the effects of economic policy. Monetary policy, fiscal policy, and public announcements depend in part on how households interpret economic information, yet households often form beliefs in ways that differ from the predictions of standard macroeconomic models and expert forecasts.

Inflation expectations are central to monetary policy transmission because central banks set nominal interest rates, but households' consumption and saving decisions depend on real interest rates, which are shaped by expected inflation. Yet the process through which households form these expectations remains imperfectly understood. A large literature has shown that households place substantial weight on salient prices, especially gasoline and food, when forming beliefs about inflation (e.g., Coibion and Gorodnichenko, 2015; D'Acunto et al., 2021). However, much of the existing evidence is based on correlations between gasoline prices and inflation expectations, making it difficult to isolate the causal effect of salient prices on household beliefs.

In joint work with Ben Klopock (Texas A&M), “[4] **Gas Tax Holiday**,” we provide quasi-experimental evidence that gasoline prices causally shift household inflation expectations. We exploit temporary state-level gasoline tax suspensions during 2022 to isolate policy-induced variation in retail gasoline prices. This setting allows us to distinguish the effect of gasoline prices from broader macroeconomic shocks, such as changes in monetary policy, supply-chain disruptions, or global oil market shocks.

We show that households revise their inflation expectations downward when gasoline prices fall, and that the magnitude of this response depends on the degree to which gas tax cuts are passed through to retail prices. The effects are economically large and substantially larger than those implied by earlier estimates based on time-series variation in energy prices. These findings provide direct causal evidence that households place disproportionate weight on gasoline prices when forming beliefs about aggregate inflation. They are also policy-relevant because gas tax holidays are often proposed as a way to provide relief when gasoline prices rise, yet their broader effects depend on both actual pass-through to retail prices and households' awareness of the policy. The paper was presented at multiple venues, including NBER and the Federal Reserve Board, and published in the *Journal of Monetary Economics*. It has also been cited in policy-facing research on gasoline prices and inflation expectations, including an Economic Bulletin article by the Federal Reserve Bank of Kansas City.

Another set of projects turns to how households interpret policy communication. I view communication as a two-sided process: policy messages must be interpreted by receivers, but they are also shaped by the incentives, constraints, and preferences of senders.

In “[5] **Liberation Day**,” joint with Jorge Hirs-Garzón (Texas A&M) and Taewon Hwang (Texas A&M), we study how Americans interpreted the April 2025 U.S. tariff announcement and whether research-based information can alter those beliefs. Using a survey experiment fielded shortly after the

announcement, we show that Americans hold heterogeneous subjective models of how tariffs affect inflation, unemployment, and the stock market. Demographic characteristics explain relatively little of this variation, while political affiliation plays a central role.

We then randomly provided respondents with summaries of academic research on the 2018–2019 U.S.–China tariff episode, covering price pass-through, employment effects, and stock market responses. We find that research-based information can shift expectations, but the form of updating depends on prior beliefs and partisan identities. When respondents broadly agree on the direction of an effect, as with inflation, information mainly changes the perceived magnitude of the effect. When prior beliefs are more dispersed, as with unemployment, information can shift respondents' expectations toward the economic mechanism emphasized in the research evidence provided in the treatment.

This paper contributes to the literature on polarized expectations by shifting the focus from monetary policy (Kuang et al. (2024, 2025), Binder et al. (2025)) to a salient real-world trade policy shock. It also contributes to the literature on subjective models of the economy by showing that heterogeneity in economic beliefs reflects partisan identities. In many hypothetical survey scenarios studied in earlier work, partisan expectations are difficult to examine because the policy under consideration is not clearly associated with a particular politician or political party. By focusing on a real-world policy announcement, we are able to observe how individuals interpret a policy that is closely linked to a prominent political figure and partisan debate. Our results show that political identity shapes how households understand inflation, unemployment, and financial markets. The patterns of belief updating after the information treatment highlight the importance of communication in policy transmission and suggest that political polarization can limit the ability of research-based information to generate a shared understanding of economic policy. I was invited to present this paper in the CEPR Central Bank Communication Seminar Series and at the Bank of Korea, and I am scheduled to present it at the Bank of Finland, reflecting its relevance to the growing literature on political polarization and policy communication.

This receiver-side evidence naturally raises a complementary question: how do information senders choose what to communicate and how to frame it? In modern macroeconomics, policy communication is not simply a way of describing policy actions; it is itself a policy tool. Central banks and other policymakers routinely communicate to shape expectations, maintain credibility, and influence household and market behavior. Understanding the behavioral preferences and constraints on such strategic communication is therefore important for evaluating the credibility and effectiveness of policy. In “[6] **Deception Aversion**,” with Béla Elmschauser and Evan Friedman, we study one such constraint: individuals' aversion to deceiving others. We develop a novel framework for measuring deception aversion without observing higher-order beliefs and find that approximately 30 percent of participants exhibit deception-averse preferences in a laboratory experiment. This work contributes to the behavioral foundations of strategic communication and complements my research on expectation formation by examining the sender side of communication rather than the receiver side. This paper is scheduled to be presented at the Stanford SITE Conference.

Together, these projects demonstrate that policy transmission depends not only on economic fundamentals, but also on how information is communicated, interpreted, and incorporated into beliefs. They also suggest that informational frictions can affect labor-market responses. My ongoing work, “[8] **Effective Labor Supply**,” joint with ChaeWon Baek (Sungkyunkwan University) and Vitaliia Yaremko (Trinity College Dublin), builds on this connection by examining how workers respond to inflation-induced declines in real wages. Using survey experiments, we study whether workers adjust their effective labor supply differently when their purchasing power falls through inflation rather than through explicit nominal wage cuts. This project extends my broader research agenda by linking expectation formation to labor-market behavior and connecting my work on inflation expectations with my research on wage rigidity. Looking ahead, I plan to continue linking expectation formation to labor-market behavior through projects on how beliefs about careers, wages, and workplace flexibility shape labor supply and fertility decisions using survey and field experiments.

Research Papers

[1] [Beyond the Spike at Zero: Understanding Nominal Wage Rigidity through Empirical and Model-Based Approaches](#). *Journal of International Economics*, 157, 104117 (2025). Referred to as “[1] Spike at Zero.”

[2] [State Dependent Government Spending Multipliers: Downward Nominal Wage Rigidity and Sources of Business Cycle Fluctuations](#), with Sarah Zubairy. *American Economic Journal: Macroeconomics*, 17(1), 379–413 (2025). Referred to as “[2] Multiplier.”

[3] [Revisiting Wage Rigidity and Welfare in Currency Unions: The Role of Downward Nominal Wage Rigidity](#), with Joonseok Oh. Revised and Resubmitted, *Journal of Money, Credit and Banking*. Referred to as “[3] Currency Union.”

[4] [Fueling Expectations: The Causal Impact of Gas Prices on Inflation Expectations](#), with Ben Klopach. *Journal of Monetary Economics*, 153, 103797 (2025). Referred to as “[4] Gas Tax Holiday.”

[5] [How Partisanship Shapes Economic Expectations: Evidence from the 2025 U.S. Tariff Announcement](#), with Jorge Hirs Garzón and Taewon Hwang, Under Review. Referred to as “[5] Liberation Day.”

[6] [Deception Aversion](#), with Béla Elmschauser and Evan Friedman, Under Review. Referred to as “[6] Deception Aversion.”

[7] [The Impact of Retail E-Commerce on Relative Prices and Consumer Welfare](#), with Misaki Matsumura and David E. Weinstein. *Review of Economics and Statistics*, 106(6), 1675–1689 (2024). Referred to as “[7] E-Commerce.”

Selected Work in Progress

[8] Inflation and Effective Labor Supply, with ChaeWon Baek and Vitaliia Yaremko. [AEA RCT Registry 0017600](#). Referred to as “[8] Effective Labor Supply.”

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Ramey, V.A. and Zubairy, S., 2018. Government spending multipliers in good times and in bad: evidence from US historical data. *Journal of Political Economy*, 126(2), pp.850–901.

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